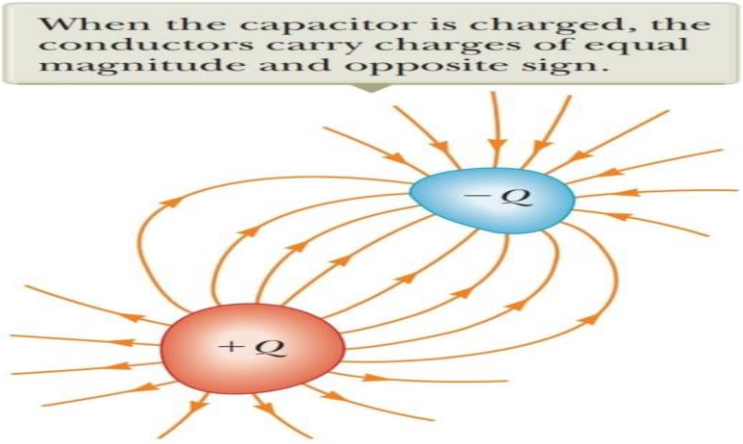
Chapter (26):- Capacitance and Dielectrics

26.1 Definition of capacitance:-

Capacitor consists of two conductors which carry charges of equal magnitude and opposite sign.

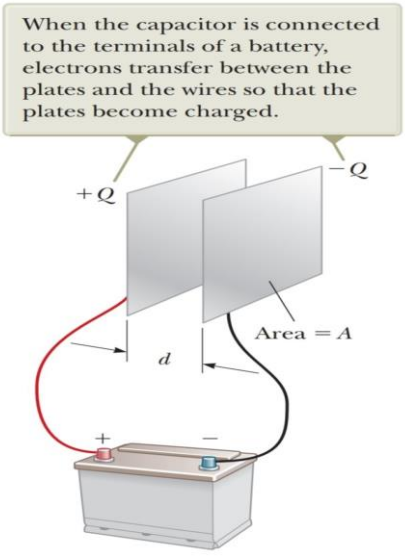


each conductor called plates. and for charged capacitors a potential difference exists between them (ΔV).

Capacitance:- the ratio of magnitude of the charge on either conductor to the magnitude of the potential difference between the conductors.

$$C = \left| \frac{Q}{\Delta V} \right| \text{ , } C \text{ is always +ve}$$

Q and ΔV expressed as +ve quantities.



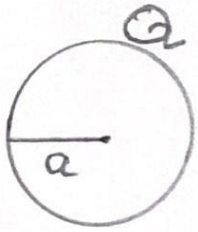
$$[C] \equiv \frac{1\,C}{1\,V} \equiv 1 \text{ Farad}$$

$$1\,\mu F = 10^{-6}\,F \quad , \quad 1\,pF = 10^{-12}\,F$$

26.2 Calculating Capacitance.

a single conductor has capacitance

① Capacitance of single spherical charged conductor :-



$$C = \frac{Q}{\Delta V}$$

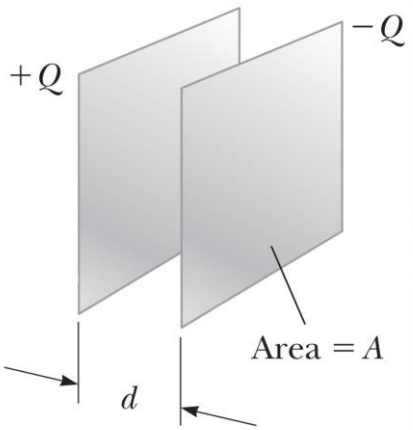
$$V = \frac{KQ}{a} \quad \text{for sphere of radius } a.$$

to find ΔV, imagin that a spherical shell of radius ∞ surronds the spherical conductor and concentric with it, it's potential is zero. $\Rightarrow \Delta V = \frac{KQ}{a} - 0 = \frac{KQ}{a}$

$$\therefore C = \frac{Q}{KQ/a} = \frac{a}{K} = 4\pi\epsilon_0 a$$

$\therefore$ for isolated charged sphere C depends on the radius directly propotional and is independent on the charge and potential.

② parallel plate capacitor:-



$$\sigma = \frac{Q}{A}$$

Electric field is uniform between the plates and zero elsewhere.

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

$$\Delta V = E d = \frac{Q d}{\epsilon_0 A}$$

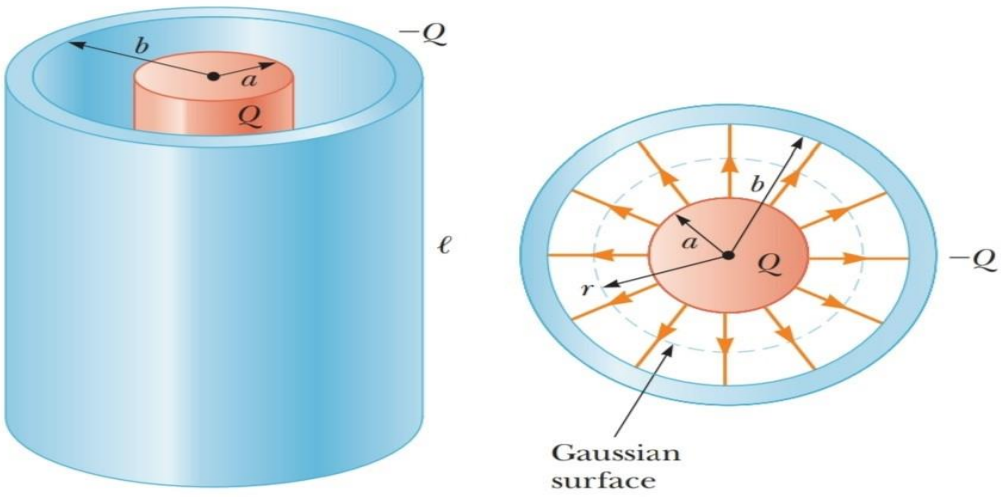
$$C = \frac{Q}{\Delta V} = \frac{Q}{Qd/\epsilon_0 A}$$

$$C = \frac{\epsilon_0 A}{d}$$

Capacitance of parallel plate capacitor is directly proportional to the area of it's plates and inversely proportional to it's separation.

$$C \propto A \qquad C \propto \frac{1}{d}$$

Ex(26.1):- The Cylindrical Capaeitor.



gaussian surface.

$$C = \frac{Q}{\Delta V}$$

$$\Delta V = V_b - V_a = -\int_a^b \vec{E}\cdot d\vec{s} = -\int_a^b E_r\, dr$$

for cylindrically charge distribution

$$E = \frac{2K\lambda}{r}$$

$$\Delta V = -2K\lambda \int_a^b \frac{dr}{r} = -2K\lambda \ln r\Big]_a^b$$

$$= -2K\lambda \ln\left(\frac{b}{a}\right)$$

$$C = \frac{Q}{2K\frac{Q}{L}\ln\left(\frac{b}{a}\right)} = \frac{L}{2K\ln(b/a)}$$

the capaeitance of cylindrical capaeitor depend on the radii a and b and proportional to the length of cylinders.

What if ?? suppose $b = 2a$, to increase the capacitance we increase a by 10%.

$$b = 2a \qquad a' = 1.1a$$

$$C' = \frac{L}{2K \ln(b/a')} \quad , \quad C = \frac{L}{2K \ln(b/a)}$$

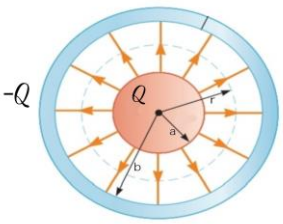
$$\frac{C'}{C} = \frac{L/2K \ln(b/a')}{L/2K \ln(b/a)} = \frac{\ln(b/a)}{\ln(b/a')}$$

$$= \frac{\ln(2a/a)}{\ln(2a/1.1a)} = \frac{\ln 2}{\ln 1.82} = 1.16$$

16 % increase in capacitance.

Ex(26.2):- The spherical capacitor :-

$$\Delta V = V_b - V_a = -\int_a^b \vec{E} \cdot d\vec{s} = -\int_a^b E_r dr \quad -(1)$$



for spherical conductor

$$E = \frac{KQ}{r^2}$$

$$\Delta V = -\int_a^b KQ \frac{dr}{r^2} = \frac{KQ}{r}\Big]_a^b$$

$$= KQ\left(\frac{1}{b} - \frac{1}{a}\right) = KQ\frac{(a-b)}{ab}$$

$$C = \frac{Q}{\Delta V} = \frac{Q\,ab}{KQ(a-b)}$$

$$= \frac{ab}{K(a-b)}$$, C depends on a and b

what if the radius $b \to \infty$

$$C = \lim_{b\to\infty} \frac{ab}{K(a-b)} = \frac{ab}{Kb}$$

$$C = \frac{a}{K} = 4\pi\epsilon_0 a$$

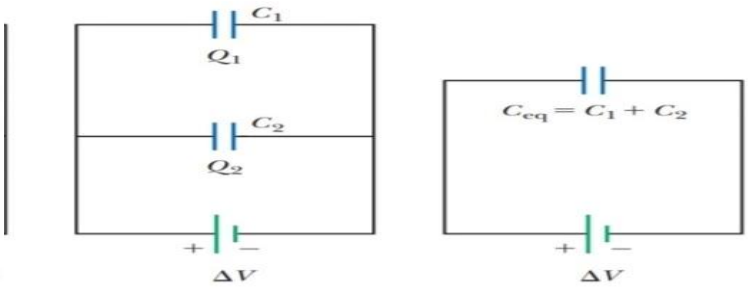
which is the capacitance of isolated conducting sphere.

26.3 Combinations of Capacitors

1- Parallel combination:-

A circuit diagram showing the two capacitors connected in parallel to a battery

A circuit diagram showing the equivalent capacitance of the capacitors in parallel



the potential difference across capacitors connected in parallel are the same and equal to potential difference applied across the combination.

$$\Delta V_1 = \Delta V_2 = \Delta V$$

$$Q_{tot} = Q_1 + Q_2$$

$$C_{eq}\,\Delta V = C_1 \Delta V_1 + C_2 \Delta V_2$$

$$C_{eq} = C_1 + C_2$$

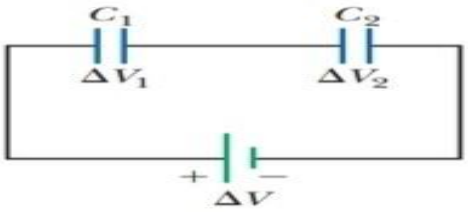
for three or more capacitors connected in Parallel $C_{eq} = C_1 + C_2 + C_3 + \cdots$

C_{eq} is greater than any of individual capacitances.

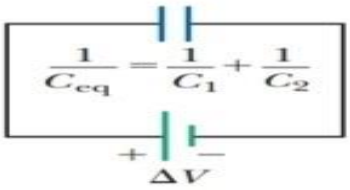
$$C_{eq} > C_1 \quad , \quad C_{eq} > C_2$$

2- Series Combination :-

A circuit diagram showing the two capacitors connected in series to a battery



A circuit diagram showing the equivalent capacitance of the capacitors in series



the charges on capacitors connected in series are same. $Q_1 = Q_2 = Q$

$$\Delta V_{tot} = \Delta V_1 + \Delta V_2$$

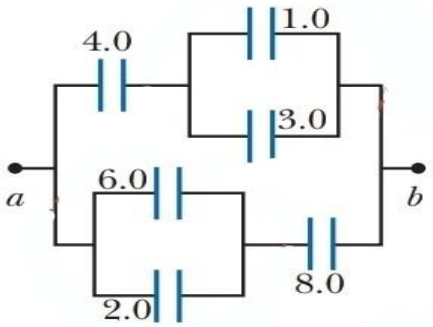
$$\frac{Q}{C_{eq}} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

For 3 or more capacitors connected in series

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$$

Ex (26.3):- Equivalent Capacitance:



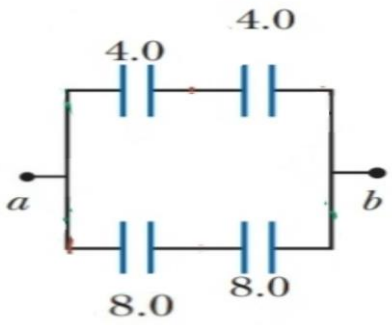
all capacitors capacitance in μf.

capacitors 1 and 3 are connected in parallel.

$C_{eq} = 1 + 3 = 4\ \mu f$

capacitors 6 and 2 are connected in parallel

$C_{eq} = 2 + 6 = 8\ \mu f$



capacitors 4 and 4 are connected in series

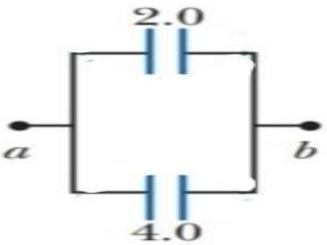
$$\frac{1}{C_{eq}} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$\therefore C_{eq} = 2\ \mu f$

Capacitors 8 and 8 are connected in series

$$\frac{1}{C_{eq}} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$C_{eq} = 4\ \mu f$



capacitors 2 and 4 are connected in parallel

$C_{eq} = 2 + 4 = 6\ \mu f$

26.4 Energy stored in a charged capacitor:-

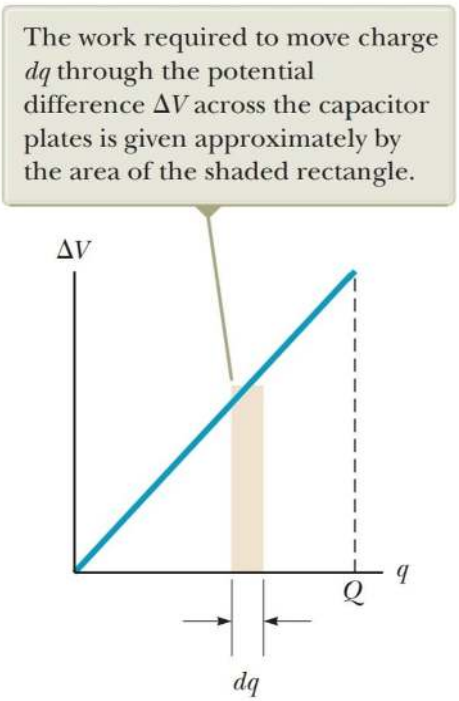
Electric potential energy is stored in the capacitor as it's transformed from the chemical energy in the battery. to calculate this energy suppose that +ve charge on one plate will move to the other plate under a force effect; work must be done on a charge to move it between two plates.

$W \simeq$ area of rectangular

$$dW = dq\,\Delta V = \frac{q}{C}\,dq$$

$$W = \int_0^Q \frac{1}{C}\,q\,dq = \frac{Q^2}{2C}$$

$$W = \Delta U \Rightarrow U = \frac{Q^2}{2C}$$



The work required to move charge dq through the potential difference ΔV across the capacitor plates is given approximately by the area of the shaded rectangle.

$$U = \frac{Q^2}{2C} = \frac{1}{2}Q\,\Delta V = \frac{1}{2}C(\Delta V)^2$$

$\equiv$ Energy stored in a charged capacitor

For parallel plate capacitor

$$C = \frac{\epsilon_0 A}{d} \quad , \quad \Delta V = Ed$$

$$U = \frac{1}{2} C (\Delta V)^2$$

$$= \frac{1}{2} \left(\frac{\epsilon_0 A}{d} \right) (Ed)^2$$

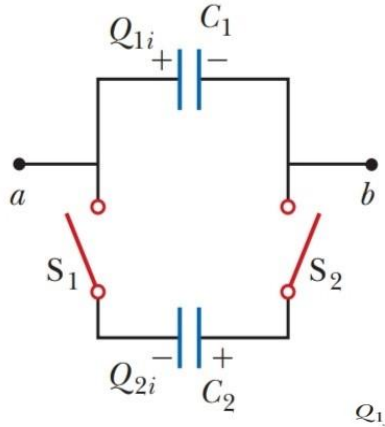
$$U = \frac{1}{2} \epsilon_0 A d E^2$$

the energy per unit volume U_E Known as energy density

$$U_E = \frac{U}{\text{Volume}} = \frac{U}{Ad} = \frac{1}{2} \epsilon_0 E^2$$

$$[U_E] \equiv J/m^3$$

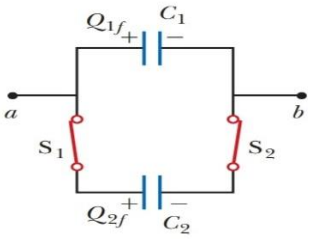
Ex (26.4):- Rewiring two charged capacitors :-



$C_1 > C_2$

$Q_{1i} > Q_{2i}$

when the switches are closed the charges redistribute. and both capacitors will have the same ΔV.



the left hand plates of the capaeitors form an isolated system because they are not connected to the right hand side.

Q_i :- is the initial charge on the left hand plates.

$$Q_i = Q_{1i} + Q_{2i}$$
$$= C_1 \Delta V_i - C_2 \Delta V_i$$
$$= (C_1 - C_2)\, \Delta V_i$$

Q_f :- is the final charge on the left hand Plates.

$$Q_f = Q_{1f} + Q_{2f} = C_1 \Delta V_f + C_2 \Delta V_f$$
$$= (C_1 + C_2)\, \Delta V_f$$

because the system is isolated $Q_f = Q_i$

$$(C_1 - C_2)\, \Delta V_i = (C_1 + C_2)\, \Delta V_f$$
$$\Delta V_f = \frac{(C_1 - C_2)}{(C_1 + C_2)}\, \Delta V_i$$

b- $U_i = \frac{1}{2} C_1 (\Delta V_i)^2 + \frac{1}{2} C_2 (\Delta V_i)^2$

$$= \frac{1}{2} (C_1 + C_2)\, \Delta V_i^2$$

$$U_f = \frac{1}{2} C_1 (\Delta V_f)^2 + \frac{1}{2} C_2 (\Delta V_f)^2$$

$$= \frac{1}{2} (C_1 + C_2) (\Delta V_f)^2$$

$$= \frac{1}{2} (C_1 + C_2) \left[\frac{C_1 - C_2}{C_1 + C_2} \right]^2 \Delta V_i^2$$

$$= \frac{1}{2} \frac{(C_1 - C_2)^2}{C_1 + C_2} \Delta V_i^2$$

$$\frac{U_f}{U_i} = \frac{\frac{1}{2} \frac{(C_1 - C_2)^2}{C_1 + C_2} \Delta V_i^2}{\frac{1}{2} (C_1 + C_2) \Delta V_i^2}$$

$$\frac{U_f}{U_i} = \frac{(C_1 - C_2)^2}{(C_1 + C_2)^2} < 1$$

$U_f < U_i$; the missing energy transfered out of the system by the mechanism of electromagnetic waves.

what if ? if $C_1 = C_2$

$Q_{i1} = Q_{2i}$ and

$Q_{1f} = Q_{f2} = 0$ in opposite Polarity.

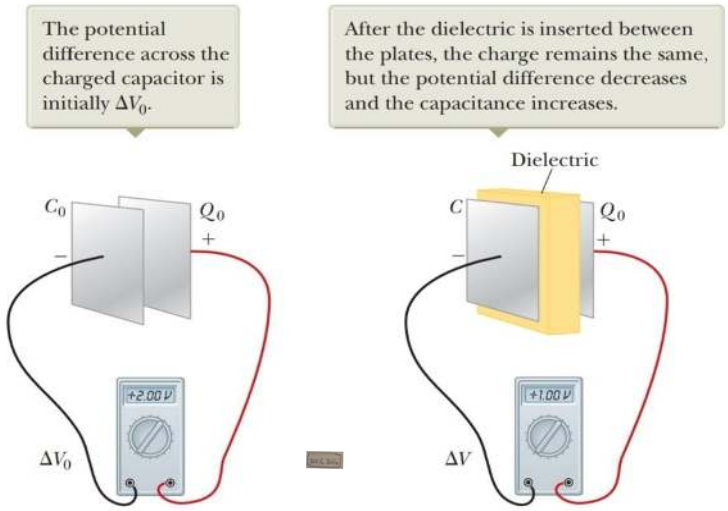
26.5 Capacitors with dielectrics:-

dielectric is anonconducting material.

by experiments they found that potential differenc across capaeitor decreases by inserting dielectric between capaeitor terminals.

for capaeitor without dielectric $\Delta V_o = \frac{Q_o}{C_o}$

where ΔV_o is the potential difference between capaeitor terminals, Q_o is the charge on capaeitor and C_o is the capaeitor.



Dielectri as NO charge can flow through voltmeter and NO battery in the circuits; this means that Q_o will not change with and without dieleetric.

the voltage with and without dielectric are relateed by dielectric constant (K).

$$\Delta V = \frac{\Delta V_o}{K}$$

$\Delta V < \Delta V_o \Rightarrow K > 1$, K is dimensionless constant.

$$C = \frac{Q_o}{\Delta V} = \frac{Q_o}{\Delta V_o / K} = K \frac{Q_o}{C_o}$$

$$C = K C_o$$

∴ after the dielectric is inserted between the plates:- 1- the charge remains the same.

2- potential difference decreases.

3- capacitance increases.

for parallel plate capacitor :-

$$C = K C_o = K \frac{\epsilon_o A}{d}$$

we can increase the capacitance of parallel-plate capacitor by inserting dielectric and decreasing (d) between the plates.

the lowest value of (d) is limited by the electric discharge that could occur through the dielectric.

max voltage can be applied to the capacitor

depends on dielectric strength.

dielectric strength :- is the max electric field of the dielectric. and can be measured in (V/m)

if the magnitude of $\vec{E}$ in the dielectric exceeds dielectric strength the dielectric begins to conduct!

working voltage
breakdown voltage
rated voltage
] 3 names for the same Parameter.

this parameter represents the largest voltage that can be applied to the capacitor without exceeding the dielectric strength.

the value of K depends on the material of dielectric. see Table (26.1) page 791 in the text book, please.

advanges of dielectric between capacitor terminals :-

1- an increase in capacitance.

2- an increase in the max operating voltage.

3- mechanical support between the plates which decrease (d) and increase (C).

Ex(26.5):- Energy stored before and after.

U_o :- energy stored in capacitor without dielectric

$$U_o = \frac{Q_o^2}{2C_o}$$

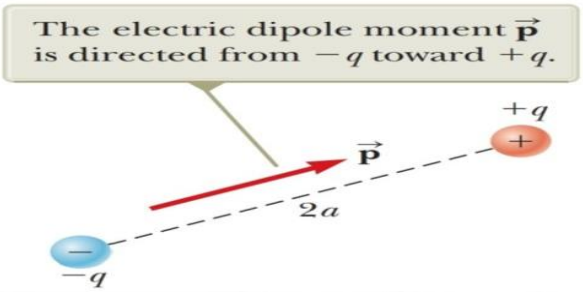
U :- energy stored in capacitour with dielectric

$$U = \frac{Q_o}{2C} = \frac{Q_o}{2C_oK} = \frac{U_o}{K}$$

$\therefore U < U_o$; energy stored in the capacitor with dielectric is less than without dielectric.

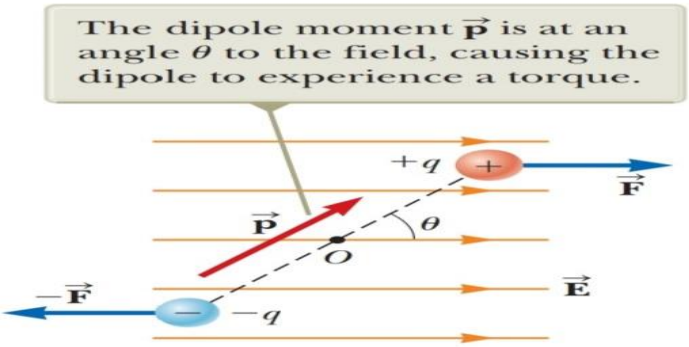
26.6 Electric dipole in an electric field :-

The electric dipole consists of 2 charges of equal magnitude and opposite sign seperated by a distance (2a).



$\vec{p}$ is the electric dipole moment which is defined as a vector directed from -ve charge to +ve charge along (2a).

For electric dipole in a uniform electric field $\vec{E}$:-



$$p = 2aq \equiv [C \cdot m]$$

the electric force acting on the two charges are equal in magnitude and opposite in direction $\Rightarrow$ the net force on the dipole = 0.

the two forces produce anet torque on the dipole.

$$\vec{\tau} = \vec{F} \times \vec{r} \equiv [N.m]$$

$$= F a \sin\theta + F a \sin\theta \quad \text{clockwise rotation}$$

$$= 2 F a \sin\theta$$

but $F = qE$, $P = 2aq$

$$\therefore \tau = 2qEa\sin\theta$$

$$= PE\sin\theta$$

$$\Rightarrow \vec{\tau} = \vec{P} \times \vec{E} = [C.m.\frac{N}{C}] = [N.m]$$

the work done by external force to rotate the dipole, the work done is stored as electric potential in the system.

the work required to rotate the dipole through an angle ($d\theta$) is:-

$$dw = \tau\, d\theta$$

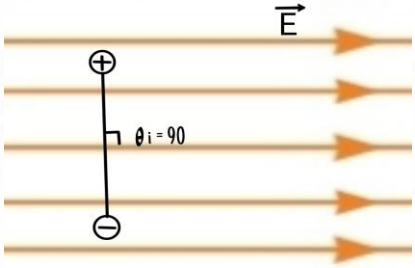
$$w = \int_{\theta_i}^{\theta_f} \tau\, d\theta = \Delta U$$

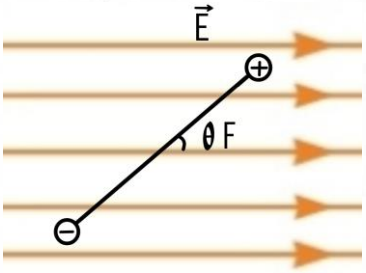
$$U_f - U_i = \int_{\theta_i}^{\theta_f} PE \sin\theta \, d\theta$$

$$= PE \int_{\theta_i}^{\theta_f} \sin\theta \, d\theta$$

$$= PE \left(-\cos\theta\right)_{\theta_i}^{\theta_f}$$

$$= PE \left(\cos\theta_i - \cos\theta_f\right) \equiv [J]$$





let $\theta_i = 90^\circ \Rightarrow U_i = 0$

$$U_f - 0 = PE\left(0 - \cos\theta_f\right)$$

$$U_f = -PE\cos\theta_f$$

$$U_E = -\vec{P}\cdot\vec{E}$$

$U_g = mgy$
$U_E = -PE\cos\theta$

1- both expressions contain parameters of the entity placed in the field mass and dipole moment

2- both expressions contain field g and E.

3- both expressions contain configuration description translational (y) and rotational (θ).

Ex(26.6):- The H_2O molecule:-

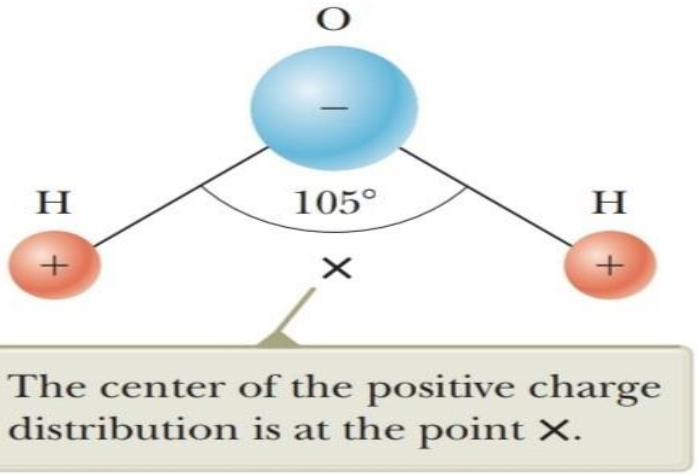
for H_2O

$P = 6.3 \times 10^{-30}$ C.m

$E = 2.5 \times 10^{5}$ N/C

$N = 10^{21}$ molecules

$\theta_i = 0 \quad , \theta_f = 90^\circ$



$$W = \Delta U_E = U_{90} - U_0$$

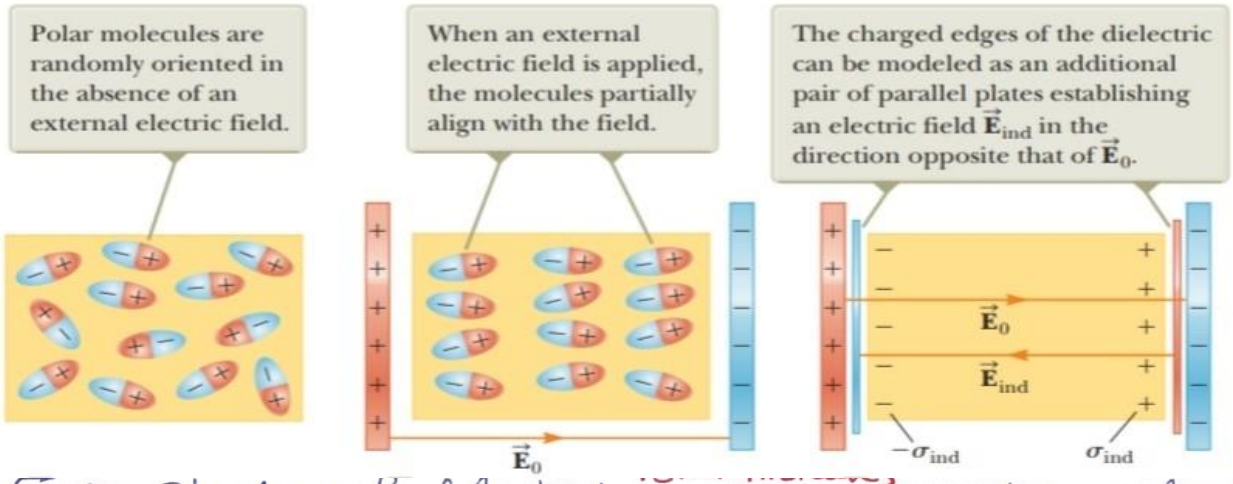
$$= [-NPE\cos 90] - [-NPE\cos 0]$$

$$= NPE$$

$$= 10^{21} \times 6.3 \times 10^{-30} \times 2.5 \times 10^{5}$$

$$= 1.6 \times 10^{-3} \text{ J}$$

26.7 An atomic description of dielectrics:-



E_0 :- Electric field between capacitor terminals without dielectric.

dielectric made of polar molcules, when an electric field is applied, the molcules partially align with the field.

$\vec{E}'$:- induced electric field due to induced charge on dielectric.

$\vec{E}_{tot} = \vec{E}_0 - \vec{E}'$, Electric field is reduced by inserting dielectric between capacitor terminals.

$E = \frac{E_0}{K}$ and this explain why ΔV reduced also.

$\Delta V = Ed$

$= \frac{E_0 d}{K} = \frac{\Delta V_0}{K}$

surface charge density $\sigma = \frac{Q}{A} \equiv [C/m^2]$

$$E = \frac{\sigma}{\epsilon_0}$$

$$E' \equiv E_{ind} = \frac{\sigma_{ind}}{\epsilon_0} \quad , \sigma_{ind} = \frac{Q_{ind}}{\epsilon_0}$$

$$E = E_0 - E_{ind}$$

$$\frac{\sigma}{K\epsilon_0} = \frac{\sigma}{\epsilon_0} - \frac{\sigma_{ind}}{\epsilon_0}$$

$$\sigma_{ind} = \sigma - \frac{\sigma}{K} = \sigma\left(1 - \frac{1}{k}\right)$$

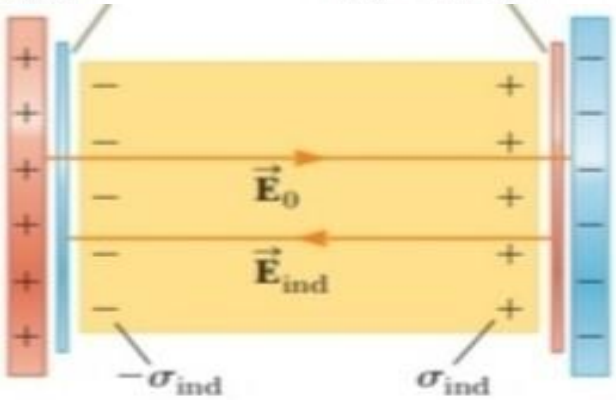
$$\sigma_{ind} = \sigma\left(\frac{K-1}{K}\right) \Rightarrow \sigma_{ind} < \sigma$$

for $K = 1$; NO dielectric between terminals $\Rightarrow$

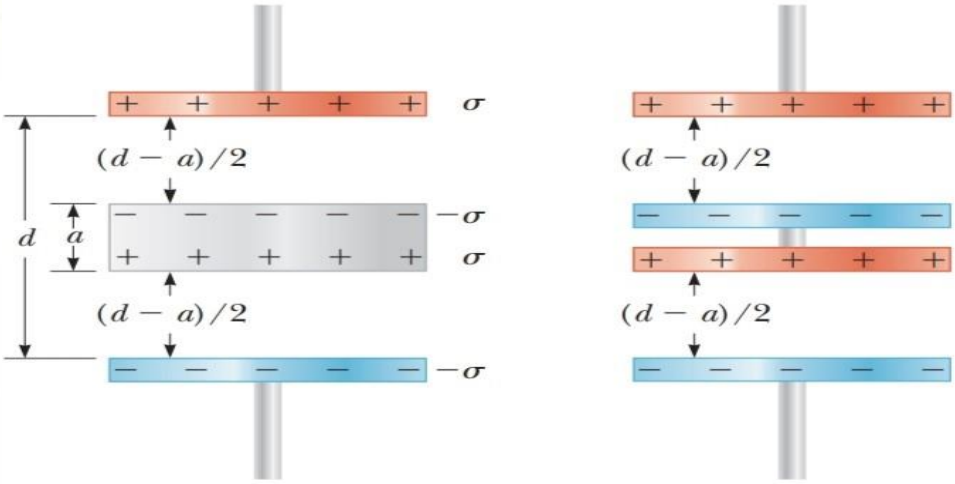
$$\sigma_{ind} = 0$$

if we put electrical conductor between conductor terminals $\Rightarrow$ $E = 0$ inside conductor $\Rightarrow$

$$E_0 = E_{ind} \Rightarrow \sigma_{ind} = \sigma$$



Ex(26.7):- Effect of metalic slab:-



inserting metalic slab between terminals of capacitor change it from one capacitor into 2 capacitors that are connected in series.

why in series not parallel ??

because $\Delta V = \Delta V_1 + \Delta V_2$

$$E \cdot d = E\frac{(d-a)}{2} + E\frac{(d-a)}{2}$$

$$a-\ \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C_1 = \frac{\epsilon_0 A}{(d-a)/2} \quad , \quad C_2 = \frac{\epsilon_0 A}{(d-a)/2}$$

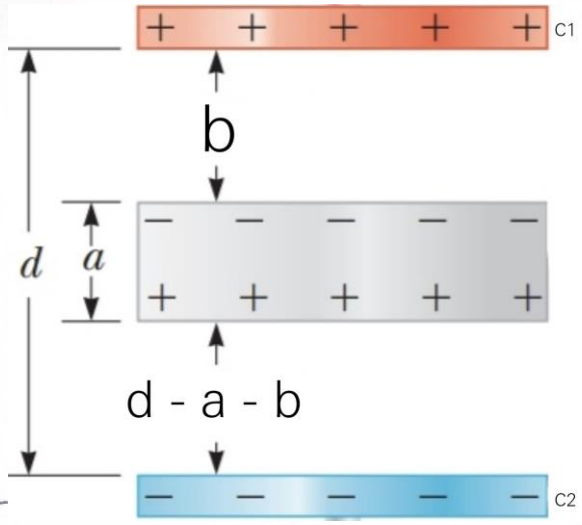
$$\frac{1}{C} = \frac{d-a}{2\epsilon_0 A} + \frac{d-a}{2\epsilon_0 A} = \frac{d-a}{\epsilon_0 A}$$

$$\boxed{C = \frac{\epsilon_0 A}{d-a}}$$

b- for thin slab, $a \to 0$

$$C = \frac{\epsilon_0 A}{d-0} = \frac{\epsilon_0 A}{d}$$

C- what if metalic slab not midway between the plates?



$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

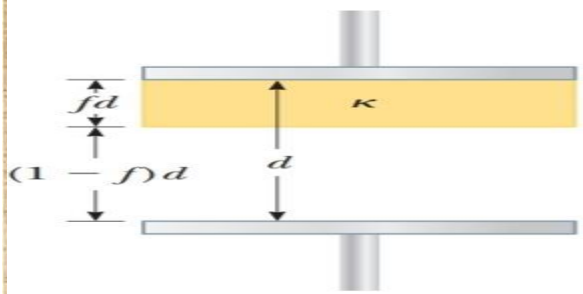
$$= \frac{1}{\frac{\epsilon_0 A}{b}} + \frac{1}{\frac{\epsilon_0 A}{d-a-b}}$$

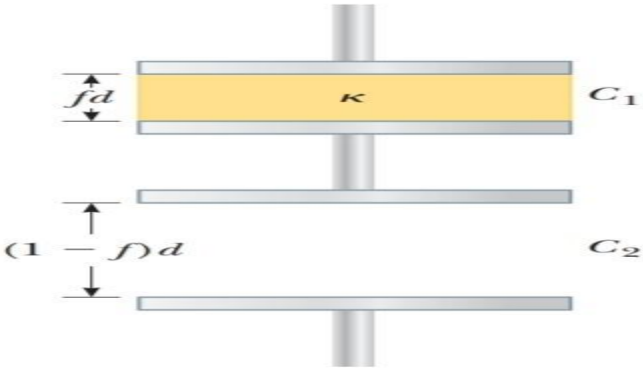
$$= \frac{b}{\epsilon_0 A} + \frac{d-a-b}{\epsilon_0 A}$$

$$\frac{1}{C} = \frac{d-a}{\epsilon_0 A} \Rightarrow C = \frac{\epsilon_0 A}{d-a}$$

∴ total capacitance does not affect by the position of metalic slab.

Ex(26.8):- Partially filled capacitor:-





f is fraction between (0-1)

$$C_1 = \frac{K\epsilon_0 A}{fd}$$

$$C_2 = \frac{\epsilon_0 A}{(1-f)d}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

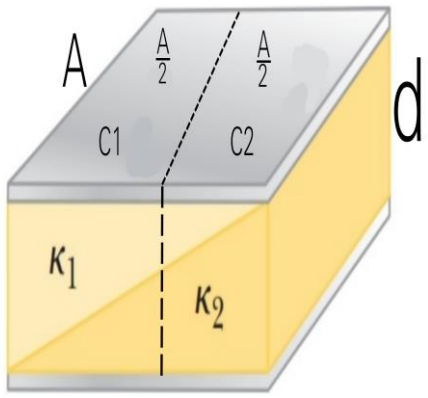
$$= \frac{fd}{K\epsilon_0 A} + \frac{(1-f)d}{\epsilon_0 A}$$

$$= \frac{fd}{K\epsilon_0 A} + \frac{K(1-f)d}{K\epsilon_0 A}$$

$$= \frac{d}{\epsilon_0 A}\left[\frac{f + K(1-f)}{K}\right]$$

$$C = \frac{\epsilon_0 A}{d}\left[\frac{K}{f + K(1-f)}\right]$$

$$C = \frac{K}{f + K(1-f)}\, C_0$$



EX:- ΔV for each half is the same, we may treat the system as (2) capacitors in parallel

$$C_1 = \frac{K_1 \epsilon_0 (A/2)}{d}, \quad C_2 = \frac{K_2 \epsilon_0 (A/2)}{d}$$

$$C = C_1 + C_2 = \frac{K_1 \epsilon_0 (A/2)}{d} + \frac{K_2 \epsilon_0 (A/2)}{d}$$

$$C = \frac{\epsilon_0 A}{2d}(K_1 + K_2)$$

$$C = \frac{(K_1 + K_2)}{2} C_0$$

A parallel-plate capacitor with a plate separation d has a capacitance C_0 in the absence of a dielectric. What is the capacitance when a slab of dielectric material of dielectric constant κ and thickness $\frac{1}{3}d$ is inserted between the plates (Fig. 26.29a)?

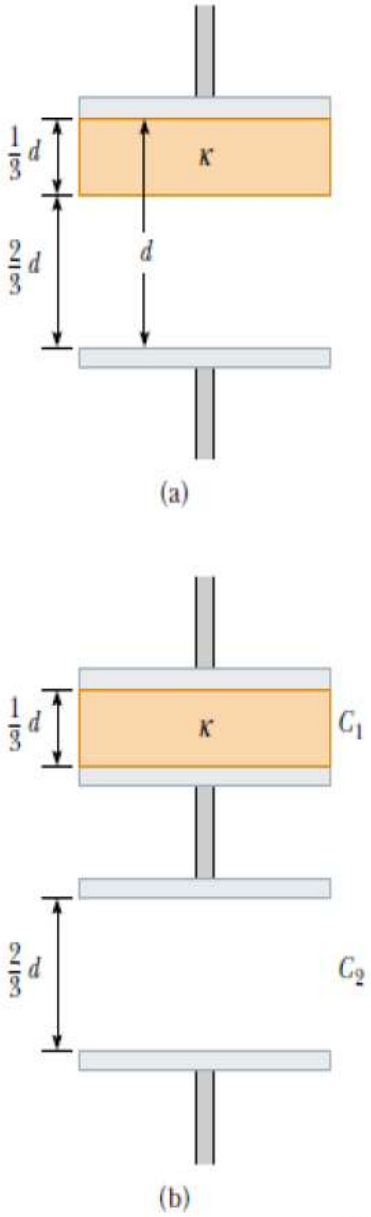


Figure 26.29 (Example 26.10) (a) A parallel-plate capacitor of plate separation d partially filled with a dielectric of thickness $d/3$. (b) The equivalent circuit of the capacitor consists of two capacitors connected in series.

$$C_1 = \frac{\kappa\epsilon_0 A}{d/3} \quad \text{and} \quad C_2 = \frac{\epsilon_0 A}{2d/3}$$

Using Equation 26.10 for two capacitors combined in series, we have

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d/3}{\kappa\epsilon_0 A} + \frac{2d/3}{\epsilon_0 A}$$

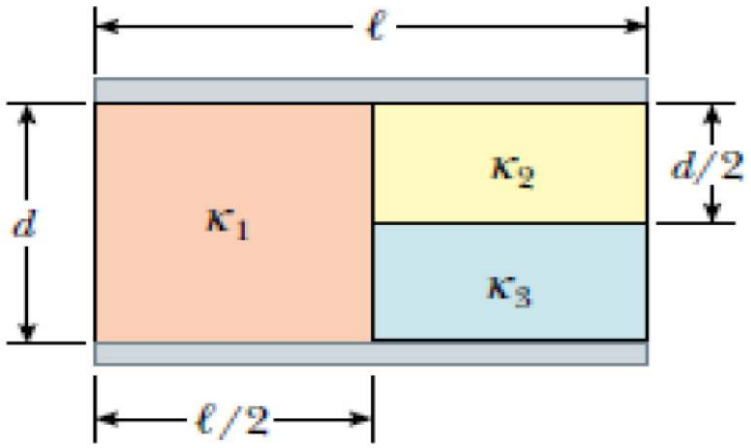
$$\frac{1}{C} = \frac{d}{3\epsilon_0 A}\left(\frac{1}{\kappa} + 2\right) = \frac{d}{3\epsilon_0 A}\left(\frac{1+2\kappa}{\kappa}\right)$$

$$C = \left(\frac{3\kappa}{2\kappa+1}\right)\frac{\epsilon_0 A}{d}$$

Because the capacitance without the dielectric is $C_0 = \epsilon_0 A/d$, we see that

$$C = \left(\frac{3\kappa}{2\kappa+1}\right)C_0$$

A parallel-plate capacitor is constructed by filling the space between two square plates with blocks of three dielectric materials, as in Figure P26.61. You may assume that $\ell \gg d$. (a) Find an expression for the capacitance of the device in terms of the plate area A and d, κ_1, κ_2, and κ_3. (b) Calculate the capacitance using the values $A = 1.00\ \text{cm}^2$, $d = 2.00\ \text{mm}$, $\kappa_1 = 4.90$, $\kappa_2 = 5.60$, and $\kappa_3 = 2.10$.



(a) $$C_1 = \frac{\kappa_1 \epsilon_0 A/2}{d};\ C_2 = \frac{\kappa_2 \epsilon_0 A/2}{d/2};\ C_3 = \frac{\kappa_3 \epsilon_0 A/2}{d/2}$$

$$\left(\frac{1}{C_2} + \frac{1}{C_3}\right)^{-1} = \frac{C_2 C_3}{C_2 + C_3} = \frac{\epsilon_0 A}{d}\left(\frac{\kappa_2 \kappa_3}{\kappa_2 + \kappa_3}\right)$$

$$C = C_1 + \left(\frac{1}{C_2} + \frac{1}{C_3}\right)^{-1} = \boxed{\frac{\epsilon_0 A}{d}\left(\frac{\kappa_1}{2} + \frac{\kappa_2 \kappa_3}{\kappa_2 + \kappa_3}\right)}$$

(b) Using the given values we find: $C_{\text{total}} = 1.76 \times 10^{-12}\ \text{F} = \boxed{1.76\ \text{pF}}$